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Inventor(s)	Akbari; Payam et al.

Vehicle having drone as an off road assistant

Abstract

A system and method including an aerial drone associated with a vehicle that can serve as a surveillance tool for previewing an off-road trail that may be travelled by the vehicle. The vehicle may include a vehicle electronic control unit and drone docking station that are configured to communicate with and control the drone during surveillance flight operations.

Inventors: Akbari; Payam (LaSalle, CA), Goodes; Adam (Rochester Hills, MI), Smith; David W (Shelby Township, MI)

Applicant: FCA US LLC (Auburn Hills, MI)

Family ID: 1000008778130

Assignee: FCA US LLC (Auburn Hills, MI)

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Primary Examiner: Mehdizadeh; Navid Z.

Assistant Examiner: Tan; Oliver

Background/Summary

FIELD

(1) The present disclosure relates to a vehicle having a drone as an off-road assistant.

BACKGROUND

(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) Many people enjoy using vehicles that are equipped to travel off-road. Indeed, there are many off-road trails throughout the United States and other countries that have various levels of difficulty. For example, some off-road trails have large rocks and bodies of water that must be traversed while in the vehicle. Typically, before various off-road trails that have various obstacles (e.g., large rocks and/or bodies of water) are driven using the vehicle, an occupant of the vehicle may first walk the off-road trail to determine whether any substantial obstacles are located along the off-road trail, which can be time-consuming. It is desirable, therefore, to provide a way to avoid having to first walk the off-road trail to determine the location of obstacles before using the vehicle to travel along the off-road trail.

SUMMARY

(4) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(5) According to a first aspect of the present disclosure, there is provided a system for controlling an aerial drone, including a vehicle body defining an interior; a vehicle electronic control unit (ECU) located within the interior of the vehicle; a drone docking station positioned within the interior of the vehicle and configured to mate with the aerial drone, and including a docking station controller configured to communicate with the vehicle ECU; and at least one of a graphic user interface (GUI) provided in the interior of the vehicle and configured to communicate with the

vehicle ECU and a remote operating device configured to communicate with the vehicle ECU, each of the GUI and remote operating device being configured for receipt of an input to conduct a surveillance flight using the aerial drone, wherein upon input of a request to either the GUI or remote operating device to conduct the surveillance flight using the drone, the GUI or remote operating device is configured to transmit a signal indicative of the request to the vehicle ECU, upon receipt of the request by the ECU, the ECU is configured to communicate an instruction to the drone docking station controller to initiate surveillance flight operations using the aerial drone, upon receipt of the instruction by the drone docking station controller to initiate surveillance flight operations, the drone docking station controller communicates an instruction to the aerial drone to initiate flight and conduct the surveillance flight operation.

(6) According to the first aspect, the surveillance flight operation surveys an off-road trail that is to be traversed using the vehicle.

(7) According to the first aspect, the system may include a network in communication with at least one of the vehicle ECU and drone docking station controller, wherein at least one of the vehicle ECU and drone docking station controller is configured to communicate with the aerial drone via the network.

(8) According to the first aspect, during the surveillance flight operations, the aerial drone is configured to generate at least one of images and video while in flight and transmit the images and video to at least one of the vehicle ECU and drone docking station via the network, and upon receipt of the images and video from the network, at least one of the vehicle ECU and drone docking station are configured to transmit the images and video to the GUI or remote operating device for display thereon.

(9) According to the first aspect, during the surveillance flight operations, the aerial drone is configured to generate at least one of images and video while in flight and save the images and video in a memory, and after conclusion of the surveillance flight operations by the aerial drone, and the aerial drone is mated with the drone docking station, the drone docking station is configured to obtain the images and video from the memory of the aerial drone and communicate the images and video to the vehicle ECU, which then communicates the images and video to either the GUI or remote operating device for display thereon.

(10) According to the first aspect, before communicating the instruction to the drone docking station controller to initiate surveillance flight operations using the aerial drone, the vehicle ECU is configured to communicate with at least one of the network and vehicle sensors to obtain information indicative of weather conditions that can impact the surveillance flight operation.

(11) According to the first aspect, based on the information indicative of the weather conditions, the vehicle ECU is configured to communicate instructions to the drone docking station that either permit the surveillance flight operations to occur or prevent the surveillance flight operations from occurring.

(12) According to a second aspect of the present disclosure, there is provided a method of using an aerial drone to conduct surveillance of an off-road trail before travelling the off-road trail using an off-road vehicle, including receiving, by either a remote operating device or a graphic user interface (GUI) of the off-road vehicle, a request to conduct a surveillance flight using the aerial drone; displaying on a screen of either the remote operating device or GUI a map that displays the off-road trail; selecting a location of interest along the off-road trail that is transmitted by either the remote operating device or GUI to a vehicle electronic control unit (ECU), the vehicle ECU being configured to communicate with the aerial drone; generating, with the vehicle ECU, an instruction to the aerial drone to conduct surveillance of the location of interest along the off-road trail.

(13) According to the second aspect, the off-road vehicle includes an interior that includes a drone docking station that is configured to mate with the aerial drone, the drone docking station including a drone docking station controller that communicates with each of the aerial drone and the vehicle ECU, and the method further comprises transmitting the instruction from the vehicle ECU to the

drone docking station controller.

(14) According to the second aspect, the method may further include communicating from the drone docking station to the aerial drone the instruction generated by the vehicle ECU.

(15) According to the second aspect, the off-road vehicle includes a plurality of sensors in communication with the vehicle ECU that are configured to generate signals indicative of weather conditions being experienced by the off-road vehicle, and before generating the instruction to conduct surveillance, the vehicle ECU is configured to determine, based on the signals indicative of the weather received from the plurality of sensors, whether the weather conditions are satisfactory for conducting the surveillance of the off-road trail.

(16) According to the second aspect, the vehicle ECU is configured to communicate with a network and obtain information indicative of the weather conditions from the network, and before generating the instruction to conduct surveillance, the vehicle ECU is configured to determine, based on the information indicative of the weather received from the network, whether the weather conditions are satisfactory for conducting the surveillance of the off-road trail.

(17) According to the second aspect, the vehicle ECU and aerial drone are each in communication with a network, and during the surveillance of the off-road trail using the aerial drone, the aerial drone is configured to transmit images of the off-road trail to the vehicle ECU via the network, and wherein the method further comprises transmitting the images received by the vehicle ECU to either the remote operating device or GUI for viewing on the screen.

(18) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.

(2) FIG. 1 is an overhead view of a vehicle according to a principle of the present disclosure;

(3) FIG. 2 is an overhead view of an interior of the vehicle illustrated in FIG. 1;

(4) FIG. 3 is a perspective view of an example drone docking station having a drone mounted thereto, located within the interior of the vehicle illustrated in FIG. 2;

(5) FIG. 4 is a schematic diagram depicting a vehicle system including a drone docking system according to a principle of the present disclosure;

(6) FIG. 5 is a flowchart depicting an example method for controlling a drone using the drone docking system of the present disclosure;

(7) FIG. 6 is a flowchart depicting another example method for controlling a drone using the drone docking system of the present disclosure;

(8) FIG. 7 is a flowchart depicting another example method for controlling a drone using the drone docking system of the present disclosure; and

(9) FIGS. 8A and 8B are flowcharts depicting an example method of using a drone to pre-survey a travel path of the vehicle illustrated in FIG. 1.

(10) Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(11) Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific

details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

(12) FIG. 1 illustrates an example vehicle **100** that may include a drone **302** and drone docking station **304** (FIG. 3) according to the present disclosure. Vehicle **100** has a body **102** defining a roof **104**. In the illustrated embodiment, roof **104** may define an opening **106**. Opening **106** may include, for example, a glass panel **108** defining a sun roof, a moon roof, or a panoramic roof that permits light to enter the vehicle. Panel **108** does not necessarily need to be formed of glass or a transparent material. In contrast, panel **108** may be an opaque removable roof panel. In either case, it should be understood that panel **108** is movable to provide access to an interior **200** (FIG. 2) of vehicle **100**.

(13) FIG. 2 illustrates an example interior **200** of vehicle **100**. In the illustrated example, interior **200** includes a front row of seats **202**, a second row of seats **204**, and a third row of seats **206**. It should be understood, however, that second and third rows of seats **204** and **206** may be omitted such that vehicle includes a rear cargo storage area in place thereof. Interior **200** may also include a graphic user interface (GUI) **208** on a dashboard **210** of vehicle **100** that is in communication with an electronic control unit **402** (FIG. 4) of vehicle **100**. Interior **200** may include a cargo bin **211** located proximate GUI **208** between the seats **212** of first row **202**.

(14) As best shown in FIG. 3, cargo bin **211** may have a movable lid **306** that may be opened to expose and provide access to a storage area **308** of cargo bin **202**, and closed to enclose and prevent access to storage area **308**, as is known in the art. In accordance with the present disclosure, storage area **308** may include a docking station **304** configured to mate with a drone **302**. As will be described in more detail later, lid **306** may be opened to permit drone **302** to detach from docking station **304** and exit the storage area **308** while being controlled by an operator. Alternatively, the operator may simply open lid **306**, manually detach drone **302** from docking station **304**, manually remove drone **302** from vehicle **100**, and then operate drone **302**. If operator elects to control drone **302** remotely to exit storage area **308** and vehicle **100**, the operator may first open or remove panel **108** located on roof **104** of vehicle **100** to permit drone **302** to exit vehicle **100** via flight.

(15) Now referring to FIG. 4, a schematic representation of a vehicle system **400** for controlling drone **302** that includes drone docking station **304** is illustrated. In addition to docking station **304**, system **400** may include ECU **402**, GUI **208**, a communication network **404**, various vehicle sensors **406**, and an optional remote operating device **408**. Example remote operating devices **408** include remote controllers, mobile phones, portable or desktop computers, electronic tablets, and the like.

(16) ECU **402** includes a communication module **410** for communicating with various modules located throughout vehicle **100** that control various features of the vehicle such as, for example, climate control modules (not shown), vehicle exterior and interior lighting modules (not shown), engine control modules (not shown), window control modules (not shown), and the like. Communication module **410** also enables ECU **402** to communicate with GUI **208**, docking station **304**, and network **404**. ECU **402** also includes a memory **411** such as a RAM or ROM, and a controller **412** that communicates with communication module **410**, wherein controller **412** is configured to communicate various instructions to GUI **208** and docking station **304** (as well as other vehicle modules like those described above) via communication module **410**.

(17) ECU **402** is also in communication with vehicle sensors **406**. Example vehicle sensors **406** that may communicate with ECU **402** include interior and exterior temperature sensors (not shown), rain sensors (not shown), exhaust gas sensors (not shown), engine load sensors (not shown), oxygen sensors (not shown), fuel level sensors (not shown), and other vehicle sensors known to one skilled in the art. Specific sensors **406** that may communicate with ECU **402** to assist with operating drone **302** include rain sensors that are configured to transmit signals indicative of

whether vehicle **100** is being exposed to rain, velocity sensors that are configured to transmit signals indicative of a velocity of the vehicle **100**, wiper motor sensors that are configured to transmit signals indicative of the vehicle windshield wiper blades (not shown) being operated, and accelerometer or gyro-sensors that are configured to transmit signals indicative of an acceleration of the vehicle **100** or whether the vehicle **100** is located on a slope (i.e., not located on flat ground). These sensors **406** are important because these sensors **406** can generate signals that assist in determining whether to deploy or recall drone **302**, or assist in operating drone **302** when it enters or exits vehicle **100**.

(18) For example, rain sensors can communicate signals indicative of whether it is raining. Operation of drone **302** may not be desirable in adverse weather conditions. Thus, based on a signal indicative of whether it is raining transmitted by the sensor **406**, ECU **402** may be configured to send an instruction to docking station **304** that either prevents drone **302** from being deployed in the adverse weather conditions, or send an instruction to docking station **304** to recall a deployed drone **302** back to the docking station **304** so as to avoid drone **302** from being operated and/or damaged by the adverse weather conditions. Windshield wiper motor sensors **406** may be used in the same manner. These operations will be described in more detail later.

(19) It also may not be desirable to deploy drone **302** from vehicle **100**, or recall drone **302** back to vehicle **100**, when vehicle **100** is in motion. Accordingly, signals indicative of the velocity of vehicle **100** or an acceleration of vehicle **100** that are transmitted to ECU **402** can be used to assist in operation of drone **302**. For example, if ECU **402** receives a signal indicative of the velocity or acceleration of the vehicle **100**, ECU **402** may communicate an instruction to docking station **304** that prevents drone **302** from being deployed from docking station **304** while the vehicle **100** is in motion. Conversely, if drone **302** is already deployed outside of vehicle **100** and the vehicle **100** is in motion, ECU **402** can send an instruction to docking station **304**, which is subsequently communicated to drone **302**, that prevents drone **302** from returning to docking station **304**. In this regard, it is much more difficult to return drone **302** to the docking station **304** when the vehicle **100** is in motion and such an instruction can prevent drone **302** from attempting to dock with docking station **304** while the vehicle **100** is in motion to prevent unnecessary damage to either vehicle **100** or drone **302** in such an instance.

(20) Docking station **304** includes a docking station communication module **414** that communicates with ECU **402** and network **404**, a charging module **416** that is configured to charge a battery **418** of drone **302** when drone **302** is mated with docking station **304**, a docking station controller **420** that is configured to send instructions to drone **302** (via network **400** when drone **302** is deployed, or directly to drone **302** via communication module **414** when drone **302** is mated with docking station **304**) and to charging module **416**, and a GPS module **600**. Docking station controller **420** may send instructions to charging module **416** to begin charging drone **302** when drone **302** is mated with docking station **304**, or send instructions to charging module **416** to cease charging drone **302** upon receive of a signal indicative that battery **418** of drone **302** has fully charged.

(21) GUI **208** communicates with ECU **402** and may be a touch-screen operated GUI **208**. As is known in the art, GUI **208** can be used to operate various systems of vehicle **100**. For example, an occupant of vehicle **100** can operate a climate control system or a radio of the vehicle **100** using GUI **208** by touching various icons **500** on a screen **510** of the GUI **208**. Drone **302** may be operated from GUI **208** in a similar manner. For example, if an occupant of vehicle **100** elects to deploy drone **302** from the docking station **304**, GUI **208** may include various icons **500** that may be touched to operate drone **30** as best shown in FIG. 6. For example, a “deploy drone” icon **500** may be located on GUI **208** and upon the icon **500** being touched by the occupant, a signal may be sent from GUI **208** to ECU **402**, which then sends a signal to docking station **304** to deploy the drone **302**.

(22) GUI **208** may also be configured to display information relative to drone **302** such as, for

example and without limitation, whether drone **302** is connected to docking station **304** or whether battery **418** of drone **302** is charged (e.g., by displaying a percentage of charge remaining in battery **418**). If drone **302** is in use outside of vehicle **100**, GUI **208** may also be used to control movement of drone **302** (e.g., an elevation of flight, velocity of drone **302**, and the like). Any instruction input into GUI **208** is communicated to ECU **402**, which is then communicated to docking station **304**, which is then communicated to drone **302** via network **404**. An example type of GUI **208** is a Uconnect® system.

(23) Network **404** may be a wireless communication network operating in accordance with one or more standards including, but not limited to, Wi-Fi, 3GPP (3rd Generation Partnership Project), 4GPP (4th Generation Partnership Project), 5GPP (5th Generation Partnership Project), LTE (long term evolution), LTE Advanced, RFID, IEEE 802.11, Bluetooth, AMPS (advanced mobile phone services), digital AMPS, GSM (global system for mobile communications), CDMA (code division multiple access), LMDS (local multi-point distribution systems), MMDS (multi-channel-multi-point distribution systems), V2V (vehicle-to-vehicle), V2X (vehicle-to-anything), and/or variations thereof. In the illustrated embodiment, network **404** may independently communicate with drone **302**, docking station **304**, ECU **402**, and remote operating device **408**. Network **404** may also serve as a wireless conduit of communication between these devices.

(24) Drone **302** includes a drone communication module **422**, a drone controller **424**, battery **418**, various sensors **426**, a memory **431** such as a RAM or ROM, and a GPS module **700**. Optional components include a camera **428**, a light-emitting device **430** that can be operated in a manner similar to a vehicle's headlights (e.g., to illuminate the ground beneath drone **302** during flight operations), and a depth device **432** configured to determine a depth of a puddle or other body of water. For example, depth device **432** can emit a laser, sonar, or some other type of energy that can assist in determining a depth).

(25) Drone communication module **422** is configured to receive and transmit information from network **400** to other elements of the system such as, for example, docking station **304** and remote operating device **408**. If drone **302** is mated with docking station **304**, drone communication module **422** may receive and transmit information directly to docking station control module **414**.

(26) Drone sensors **426** may be similar to vehicle sensors **406**. Example drone sensors **426** include an altimeter that generates a signal indicative of an altitude of the drone **302**, a GPS sensor that generates a signal indicative of a location of the drone **302** during flight operations, a velocity sensor that generates a signal indicative of a velocity of the drone **302** during flight operations, a rain sensor that generates a signal indicative of whether it is raining, a battery sensor that generates a signal indicative of a percentage of battery power remaining in battery **418**, accelerometers that generate signals indicative of an acceleration of drone **302**, and gyro-sensors that generate signals indicative of an orientation of drone **302**. These signals may be transmitted to drone controller **424**, which are then sent to communication module **422** to be transmitted to other elements of the system **400** via network **404**.

(27) If drone **302** includes a camera **428**, images or video footage captured by camera **428** may be transmitted via communication module **422** to vehicle **100** or remote operating device **408**. For example, any images or footage captured by camera **428** may be communicated to vehicle **100** (i.e., to ECU **402** via network **400**, or to docking station **304** via network **400** and then to ECU **402** from the docking station **304**) to be displayed by GUI **208** on screen **510**. Alternatively, if remote operating device **408** includes a display **510**, the images or footage captured by camera **428** transmitted to remote operating device **408** via network **404** may be displayed by remote operating device **408**. In another alternative, the images and/or video obtained by camera **428** may be stored in memory **431** to be downloaded and viewed later.

(28) Remote operating device **408** may include a remote operating device communication module **434**, a GPS module **435**, and a remote operating device controller **436**. Remote operating device communication module **434** permits remote operating device **408** to communicate with network

404 and other elements of system **400** such as drone **302**, docking station **304**, and ECU **402**. Remote operating device controller **436** may include a drone operating system software stored therein for operating drone **302**. Based on inputs received from an operator, remote operating device controller **436** may then communicate instructions to any of drone **302**, docking station **304**, and ECU **402** via remote operating device communication module **434**.

(29) For example, if someone using remote operating device **408** desires to deploy drone **302** from docking station **304**, the operator may use remote operating device **408** to send an instruction to docking station **304** via network **404**, which then communicates with drone **302** to initiate deployment of drone **302** from docking station **304** (in such a case, it is assumed that panel **108** (FIG. 1) is open or removed).

(30) In another embodiment, the instructions sent from remote operating device **408** to docking station **304** via network **404** may be further communicated to ECU **402**. Upon receipt of such an instruction by ECU **402**, ECU **402** may communicate with a vehicle sensor **406** such as, for example, a sensor **406** that generates a signal indicative of whether panel **108** is open or removed to determine whether it is safe for drone **302** to exit the vehicle **100** through the opening **106** formed in roof **104** of vehicle **100**. After ECU **402** receives a signal indicative of panel **108** being open or removed so as to not obstruct opening **106**, ECU **402** may then communicate a signal back to docking station **304** that permits drone **302** to be deployed an exit vehicle **100** through opening **106**.

(31) In other embodiments, ECU **402** may also communicate with other sensors **406** or network **404** to determine whether weather conditions are such that drone **302** may be deployed. For example, if sensor **406** is a rain sensor that communicates a signal indicative of the vehicle **100** being exposed to rain to ECU **402**, ECU **402** may transmit an instruction to docking station **304** that prevents drone **302** from being deployed. Alternatively, ECU **402** may communicate with network **404** to retrieve current weather conditions. Based on the information regarding the current weather conditions received from network **404**, ECU **402** is configured to send an instruction to docking station **304** that either permits or prevents drone **302** from deploying from docking station **304**.

(32) Drone **302** may be any type of drone known to one skilled in the art, provided that drone **302** is sized to fit within vehicle **100**. Examples of types of drones include, but are not limited to, a helicopter drone, a fixed-wing drone, a multi-rotary-wing drone, a blimp drone, a flexible-wing drone, and the like. Drone **302** may be capable of both autonomous navigation and operating via remote control.

(33) Now referring to FIGS. 5 and 6, a method of operating drone **302** will be described. As noted above, drone **302** may be operated using GUI **208** or remote operating device **408**. Description of using GUI **208** to operate drone **302** will first be described. Firstly, if docking station **304** and drone **302** are located within cargo bin **202**, lid **306** must be opened (step **1000**) by an occupant of vehicle **100** (this step may be omitted if, for example, docking station **304** is located in a rear storage area of vehicle **100** (i.e., if vehicle **100** does not include second or third rows **204**, **206** of seats and instead includes a rear storage area)).

(34) After opening lid **305**, an occupant may then use GUI **208** to initiate flight operations of drone **302** (step **1010**). At this time, an occupant of vehicle **100** may input information into GUI **208** such as, for example, a destination for drone **302** to travel, a desired flight path, a desired altitude for the flight, a time period of the flight, and the like by interacting with various icons **500** (FIG. 6) on a screen **510** of the GUI. At this time, GUI **208** communicates a signal to ECU **402** that drone **302** is to be deployed (step **1020**) and that includes the desired information input to the GUI **208** by the occupant.

(35) Upon receipt of this signal and information, ECU **402** communicates with various sensors **406** and/or network **404** to determine whether conditions are acceptable to initiate deployment of drone **302** (step **1030**). These conditions include obtaining information relative to current weather

conditions from network **404**, or obtaining signals from sensors **406** that are indicative of the weather that vehicle **100** is currently being exposed (i.e., a signal indicative of rain from rain sensors or a signal indicative of the vehicle wiper blades being operated, for example). Other conditions include whether the vehicle **100** is in motion, whether the vehicle **100** is at an orientation (i.e., on a slope) that would make it difficult to deploy drone **302** from vehicle **100**, and whether panel **108** is open or removed to permit drone **302** to exit the vehicle **100** during flight. If these conditions are satisfactory, ECU **402** may transmit an instruction (step **1040**) to docking station **304** to permit drone **302** to begin flight operations, detach from docking station **304**, and exit vehicle **100** through opening **106** provided in roof **104** of vehicle **100**.

(36) Upon receipt by docking station **304** of instructions from ECU **402**, docking station **304** communicates instructions to drone **302** to initiate flight and including the desired flight characteristics input into GUI **208** by the occupant (step **1050**). Upon receipt of these instructions, drone **302** may autonomously detach from docking station **304** and exit vehicle **100** through opening **106** (step **1060**). At this time, drone **302** may continue to operate autonomously using the desired flight characteristics input by the occupant to GUI **208** before returning to docking station **304**.

(37) Alternatively, drone **302** may autonomously detach from docking station **304** and exit vehicle **100** before operation of drone **302** is transferred back to GUI **208** (step **1070**). In such a case, drone **302** may exit vehicle **100** and reach a predetermined altitude before operation of drone **302** is transferred back to GUI **208**. Then, the occupant can use various control icons on GUI **208** to move in a particular direction, move at a desired velocity, reach a desired altitude, use camera **428**, operate light source **430**, operate depth detection device **432**, and the like (step **1080**). At this time, the instructions input into GUI **208** are communicated to ECU **402**, which then communicates the instructions to docking station **304**, which communicates the instructions to drone **302** via network **404**.

(38) If drone **302** is operating autonomously based on instructions input to GUI **208** and the desired flight time has elapsed, drone **302** may then automatically return to vehicle **100**, pass through opening **106**, and mate with docking station **304** (step **1090**). Alternatively, if drone **302** is being manually operated, in step **1090** the occupant can input instructions into GUI **208** that direct drone **302** back to vehicle **100**. The occupant can operate drone **302** to reenter vehicle **100** and mate with docking station **304**, or the occupant can operate drone **302** to return to a location exterior to vehicle **100** before manually retrieving drone **302** and manually mating drone **302** with docking station **304**. It should be understood that if vehicle **100** has moved during flight operations, a location of vehicle **100** can be transmitted to drone **302** via network **404**. In this regard, both drone **302** and docking station **304** may include a GPS module **600**, **700** that may coordinate with each other.

(39) It should be understood that during flight operations, ECU **402** may continually communicate with network **404** and/or sensors **406** to continually make a determination whether conditions are satisfactory for continuing flight operations (e.g., continually checking weather information). If for whatever reason ECU **402** determines that flight operations should cease, ECU **402** may send an instruction to docking station **304** to have drone **302** return to vehicle **100**, which is then communicated to drone **302** using network **404**. Moreover, it should be understood that information obtained from drone sensors **426** can be continuously transmitted from drone **302** to docking station **304** and ECU **402** via network **404**. If drone sensors **426** generate signals indicative of poor operation (e.g., motor issues, poor flight conditions, and the like), ECU **402** can transmit instructions to docking station **304** that are transmitted to drone **302** via network **404** to cease flight operations and return to vehicle **100**.

(40) A similar process can be used if drone **302** is being operated by remote operating device **408**. It should be understood, however, that use of remote operating device **408** enables an operator of the remote operating device **408** to deploy drone **302** from vehicle **100** if the operator is located at a

location remote from vehicle **100**. For example, referring to FIG. 7, a control process of operating drone **302** when the operator is located remote from vehicle **100** and plans to control drone **302** using remote operating device **408** is illustrated. In the process of FIG. 7, it should be understood that operator is not located in vehicle **100** when operating drone **302**.

(41) First, in step **2000**, operator may open up a software program stored on remote operating device **408** that is configured to operate drone **302**. After opening the software program, a screen **510** of the remote operating device **408** can display an interface similar to that of GUI **208** (see, e.g., FIG. 6) and, the operator may input various information such as, for example, a destination for drone **302** to travel, a desired flight path, a desired altitude for the flight, a time period of the flight, and the like by interacting with various icons **500** (step **2010**; FIG. 6). At this time, remote operating device **408** communicates a signal to either docking station **304** and/or ECU **402** via network **404** that drone **302** is to be deployed (step **2020**) and that includes the desired information input to the remote operating device **408** by the operator.

(42) Upon receipt of this signal and information, ECU **402** communicates with various sensors **406** and/or network **404** to determine whether conditions are acceptable to initiate deployment of drone **302** (step **2030**). These conditions include obtaining information relative to current weather conditions from network **404**, or obtaining signals from sensors **406** that are indicative of the weather that vehicle **100** is currently being exposed (i.e., a signal indicative of rain from rain sensors or a signal indicative of the vehicle wiper blades being operated, for example). Other conditions include whether the vehicle **100** is in motion, whether the vehicle **100** is at an orientation (i.e., on a slope) that would make it difficult to deploy drone **302** from vehicle **100**, and whether panel **108** is open or removed to permit drone **302** to exit the vehicle **100** during flight. If these conditions are satisfactory, ECU **402** may transmit an instruction (step **2040**) to docking station **304** to permit drone **302** to begin flight operations, detach from docking station **304**, and exit vehicle **100** through opening **106** provided in roof **104** of vehicle **100**. If panel **108** is obstructing opening **106**, ECU **402** can instruct a motor (not shown) that moves panel **108** to move panel **108** so as to not obstruct opening **106** (i.e., controller **402** can open the sunroof, moonroof, or panoramic roof panel).

(43) Upon receipt by docking station **304** of instructions from ECU **402**, docking station **304** communicates instructions to drone **302** to initiate flight and including the desired flight characteristics input into remote operating device **408** by the operator (step **2050**). Upon receipt of these instructions, drone **302** may autonomously detach from docking station **304** and exit vehicle **100** through opening **106** (step **2060**). At this time, drone **302** may continue to operate autonomously using the desired flight characteristics input by the operator to remote operating device **408** before returning to docking station **304**. Alternatively, drone **302** may autonomously detach from docking station **304** and exit vehicle **100** before operation of drone **302** is transferred back to remote operating device (ROD) **408** (step **2070**). In such a case, drone **302** may exit vehicle **100** and reach a predetermined altitude before operation of drone **302** is transferred back to remote operating device **408**. Then, the occupant can use various control icons on the display of remote operating device **408** to move in a particular direction, move at a desired velocity, reach a desired altitude, use camera **428**, operate light source **430**, operate depth detection device **432**, and the like (step **2080**). At this time, the instructions input into remote operating device **408** are communicated to ECU **402**, which then communicates the instructions to docking station **304**, which communicates the instructions to drone **302** via network **404**.

(44) If drone **302** is operating autonomously based on instructions input to remote operating device **408** and the desired flight time has elapsed, drone **302** may then automatically return to vehicle **100**, pass through opening **106**, and mate with docking station **304** (step **2090**). Alternatively, if drone **302** is being manually operated by remote operating device **408**, in step **2090** the operator can input instructions into remote operating device **408** that direct drone **302** back to vehicle **100**. The operator can operate drone **302** to reenter vehicle **100** and mate with docking station **304**, or

the occupant can operate drone **302** to return to the operator's location exterior to vehicle **100** before manually retrieving drone **302** and manually mating drone **302** with docking station **304** when desired. It should be understood that if vehicle **100** has moved during flight operations, a location of vehicle **100** can be transmitted to drone **302** via network **404**. In this regard, both drone **302** and docking station **304** may include a GPS module that may coordinate with each other.

(45) It should be understood that during flight operations, ECU **402** may continually communicate with network **404** and/or sensors **406** to continually make a determination whether conditions are satisfactory for continuing flight operations (e.g., continually checking weather information). If for whatever reason ECU **402** determines that flight operations should cease, ECU **402** may send an instruction to docking station **304** to have drone **302** return to vehicle **100**, which is then communicated to drone **302** using network **404**. Moreover, it should be understood that information obtained from drone sensors **426** can be continuously transmitted from drone **302** to docking station **304** and ECU **402** via network **404**. If drone sensors **426** generate signals indicative of poor operation (e.g., motor issues, poor flight conditions, and the like), ECU **402** can transmit instructions to docking station **304** that are transmitted to drone **302** via network **404** to cease flight operations and return to vehicle **100**.

(46) Now referring to FIGS. **8A** and **8B**, a method of surveying a potential travel path of the vehicle **100** is illustrated. At step **3000**, the operator initiates a survey request. This request may be input to either remote operating device **408** or GUI **208** by selecting the appropriate icon **500** on either remote operating device **408** or GUI **208**. After selecting the appropriate icon **500** to initiate the survey request, controller **436** of remote operating device **408** may communicate with GPS module **435** of remote operating device **408** to determine a location of remote operating device **408**, at which time a map indicating the location of remote operating device **408** may be displayed on screen **510** of remote operating device **408** (step **3010**).

(47) At this time, the operator may select a location remote from remote operating device **408** and vehicle **100** to survey using drone **302** (step **3020**) by, for example, touching the map displayed on display **510**. The selected location may be, for example, an off-road trail that the operator wishes to travel using vehicle **100**, or may be another location that the operator would like to survey before traveling to with vehicle **100**. In the case that the selected location is an off-road trail, this is advantageous from the standpoint that the operator can view the off-road trail using camera **428** of drone **302** to determine any potential hazards or obstacles that vehicle **100** will be required to traverse while on the off-road trail. As noted in the background, this has conventionally been accomplished by first walking the off-road trail. Use of drone **302** to first survey the off-road trail can be effective in eliminating such a step.

(48) A similar process can be used if the survey request is input to GUI **208**. After selecting the appropriate icon **500** to initiate the survey request, GUI **208** may communicate with ECU **402** that a survey request has been input to GUI **208**. If vehicle **100** is equipped with a GPS module (not shown), ECU **402** can communicate with the GPS module to receive a location of vehicle **100**, and based on the signal received from GPS module indicative of the location of vehicle **100**, ECU **402** can communicate this information to GUI **208** and GUI **208** can display a map of the location of vehicle **100** on display **510** of GUI **208**. Alternatively, ECU **402** may receive the location of vehicle **100** from docking station **304** which is equipped with GPS module **600**. Another alternative is that ECU **402** includes various off-road trail maps stored in memory **411** that may be accessed using GUI **208**. In any event, at this time, the operator may select the location remote from vehicle **100** to survey using drone (step **3020**) by, for example, touching the map displayed on the display **510** of GUI **208**.

(49) If the survey location has been selected using remote operating device **408**, the survey location may be transmitted from remote operating device **408** to ECU **402** via network **404** (step **3030**). If the survey location has been selected using GUI **208**, the survey location may be transmitted from GUI **208** to ECU **402** (step **3030**).

(50) Once the survey location has been received by ECU 402, ECU 402 communicates with sensors 406 of vehicle 100 (step 3040) to confirm that the weather is suitable for flight operations (see, e.g., FIGS. 5 and 6). In addition, although not required, ECU 402 may communicate with docking station 304 to determine whether drone 302 has sufficient battery life to complete the operation. ECU 402 may also determine whether panel 108 of vehicle 100 is open. Further, if the vehicle 100 has a cargo lid 306 configured to be electronically operated by the ECU 402, the ECU 402 can determine whether lid 306 is open to permit drone 302 to deploy from docking station 304.

(51) In step 3050, if panel 108, and optionally lid 306, are not open, ECU 402 can either communicate instructions to remote operating device 408 (via network 404) or GUI 208 to provide an indication on display 510 of these devices to the operator that panel 108 and lid 306 should be opened before flight operations can begin, or ECU 402 can send instructions to the appropriate mechanisms (not shown) that will automatically open the cargo lid 306 and panel 108 if closed.

(52) At step 3060, the ECU 402 sends the operator-selected survey instructions to the docking station 304, which then communicates the instructions received from ECU 402 to the drone 302 (step 3070).

(53) At step 3080, after receipt of the survey instructions from docking station 304 (including, for example, the survey location and desired flight conditions such as altitude, velocity, etc.), drone 302 can detach from the docking station 302 and exit vehicle 100 through opening 106 of vehicle 100.

(54) At step 3090, the drone 302 executes the survey instructions selected by the operator. As noted above, the drone 302 may execute instructions to survey a specific area such as an off-road trail to assist the operator in determining whether any obstacles such as rocks, fallen trees, bodies of water, and the like or located along off-road trail that the vehicle 100 will encounter while traveling the off-road trail. Drone 302 may use camera 428 to take video footage of the surveyed area and store the footage in a memory 431 that may be viewed on display 510 once drone 302 returns to vehicle 100 and mates with docking station 304, where docking station 304 may transmit the video footage to ECU 402, which then either sends this information directly to GUI 208 or to remote operating device 408 via network 404. Alternatively, drone 302 may use camera 428 and communicate a live video feed to either the remote operating device 408 via network 404, or to ECU 402 via network 404, which then communicates the feed directly to GUI 208. After review of the footage obtained by drone 302, the operator may become aware of any hazards or obstacles and make a decision whether to travel the off-road trail using vehicle 100. In addition, drone 302 may be instructed to use depth device 432 to a depth of any puddles or other bodies of water that need to be traversed by vehicle 100 along trail, which can be transmitted from drone 302 via network 404 to vehicle 100.

(55) At step 3100, the operator may elect to send additional survey instructions to drone 302 or, if the survey that has been conducted by drone 302 is satisfactory to the operator, the operator may elect to recall drone 302. For example, the operator may elect to manually control the flight and/or camera 428 of the drone 302 to further inspect other areas of interest that were found during the drone's 302 survey of the selected location (step 3110).

(56) At step 3120, upon concluding the additional survey operation or recalling drone 302, the drone 302 may navigate to the docking station 304 either autonomously or while being operated by the operator.

(57) At step 3130, upon reaching the vehicle 100, the drone 302 may descend through the opening 106 and mate with the docking station 304. Alternatively, drone 302 may be instructed to land exterior to vehicle 100 to be manually obtained by the operator and then manually mated with docking station 304.

(58) At step 3140, the docking station 304 communicates to the ECU 402 that the drone 302 has mated with docking station 304 and the flight operation is concluded.

(59) At step 3150, the ECU 402 may then send instructions to the appropriate mechanism (not shown) to automatically close the panel 108 (and lid 306 if so equipped). Alternatively, the ECU

402 may notify the operator that panel **108** (and optionally lid **306**) should be closed by sending an instruction to remote operating device **408** (via network) or GUI **208** to display a note to the operator on display **510**.

(60) After conclusion of operation of drone **302**, the operator may then make the decision whether or not to traverse the off-road trail using vehicle **100** based on the survey conducted by drone **302**.

(61) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A system for controlling an aerial drone, comprising: a vehicle having a body defining an interior, wherein the interior includes a front row of seats, including a driver's seat and at least one passenger seat, and a cargo bin having a movable lid to provide and prevent access to the cargo bin, and wherein the cargo bin is located between the driver's seat and the at least one passenger seat; a vehicle electronic control unit (ECU) located within the interior of the vehicle; a drone docking station positioned within the interior of the vehicle and configured to mate with the aerial drone, wherein the drone docking station is located in the cargo bin; a docking station controller located within the drone docking station and configured to communicate with the vehicle ECU; and at least one of a graphic user interface (GUI) provided in the interior of the vehicle and configured to communicate with the vehicle ECU and a remote operating device configured to communicate with the vehicle ECU, each of the GUI and remote operating device being configured for receipt of an input to conduct a surveillance flight using the aerial drone, wherein upon input of a request to either the GUI or remote operating device to conduct the surveillance flight using the drone, the GUI or remote operating device is configured to transmit a signal indicative of the request to the vehicle ECU, upon receipt of the request by the ECU, the ECU is configured to communicate an instruction to the docking station controller to initiate surveillance flight operations using the aerial drone, upon receipt of the instruction by the docking station controller to initiate surveillance flight operations, the docking station controller is configured to communicate the instruction to the aerial drone to initiate flight and conduct the surveillance flight operation.

2. The system according to claim 1, wherein the surveillance flight operation surveys an off-road trail that is to be traversed using the vehicle.

3. The system according to claim 1, further comprising a network in communication with at least one of the vehicle ECU and docking station controller, wherein at least one of the vehicle ECU and docking station controller is configured to communicate with the aerial drone via the network.

4. The system according to claim 3, wherein during the surveillance flight operations, the aerial drone is configured to generate at least one of images and video while in flight and transmit the images and video to at least one of the vehicle ECU and drone docking station via the network, and upon receipt of the images and video from the network, at least one of the vehicle ECU and drone docking station are configured to transmit the images and video to the GUI or remote operating device for display thereon.

5. The system according to claim 3, wherein during the surveillance flight operations, the aerial drone is configured to generate at least one of images and video while in flight and save the images and video in a memory, and after conclusion of the surveillance flight operations by the aerial drone, and the aerial drone is mated with the drone docking station, the drone docking station is configured to obtain the images and video from the memory of the aerial drone and communicate

the images and video to the vehicle ECU, which then communicates the images and video to either the GUI or remote operating device for display thereon.

6. The system according to claim 3, wherein before communicating the instruction to the docking station controller to initiate surveillance flight operations using the aerial drone, the vehicle ECU is configured to communicate with at least one of the network and vehicle sensors to obtain information indicative of weather conditions that can impact the surveillance flight operation.

7. The system according to claim 6, wherein based on the information indicative of the weather conditions, the vehicle ECU is configured to communicate instructions to the drone docking station that either permit the surveillance flight operations to occur or prevent the surveillance flight operations from occurring.

8. A method of using an aerial drone to conduct surveillance of an off-road trail before travelling the off-road trail using an off-road vehicle, the off-road vehicle including a vehicle body having an interior, wherein the interior includes a front row of seats, including a driver's seat and at least one passenger seat, and a cargo bin having a movable lid that opens and closes, comprising: receiving, by either a remote operating device or a graphic user interface (GUI) of the off-road vehicle, a request to conduct a surveillance flight of the off-road trail using the aerial drone; retrieving, by either the remote operating device or GUI and from a vehicle electronic control unit (ECU) having a memory, a digital map of the off-road trail stored in the vehicle ECU's memory; displaying on a screen of either the remote operating device or GUI the digital map of the off-road trail; selecting a location of interest along the displayed digital map of the off-road trail; transmitting, by either the remote operating device or GUI, a request to conduct a surveillance flight at the selected location of interest to the vehicle ECU; determining, by the vehicle ECU, an opened/closed state of the movable lid of the cargo bin; generating, by the vehicle ECU, an instruction to conduct a surveillance flight at the selected location of interest along the off-road trail; opening, by the vehicle ECU, the movable lid of the cargo bin if the determined opened/closed state of the movable lid is closed; transmitting, by the vehicle ECU, the generated instruction to conduct a surveillance flight at the selected location of interest to a docking station controller; and transmitting, by the docking station controller, the generated instruction to conduct a surveillance flight at the selected location of interest to the aerial drone.

9. The method according to claim 8, wherein the off-road vehicle includes an interior that includes a drone docking station that is configured to mate with the aerial drone, the drone docking station including the drone docking station controller that communicates with each of the aerial drone and the vehicle ECU.

10. The method according to claim 8, wherein the off-road vehicle includes a plurality of sensors in communication with the vehicle ECU that are configured to generate signals indicative of weather conditions being experienced by the off-road vehicle, and before generating the instruction to conduct surveillance, the vehicle ECU is configured to determine, based on the signals indicative of the weather received from the plurality of sensors, whether the weather conditions are satisfactory for conducting the surveillance of the off-road trail.

11. The method according to claim 10, wherein the vehicle ECU is configured to communicate with a network and obtain information indicative of the weather conditions from the network, and before generating the instruction to conduct surveillance, the vehicle ECU is configured to determine, based on the information indicative of the weather received from the network, whether the weather conditions are satisfactory for conducting the surveillance of the off-road trail.

12. The method according to claim 8, wherein the vehicle ECU and aerial drone are each in communication with a network, and during the surveillance of the off-road trail using the aerial drone, the aerial drone is configured to transmit images of the off-road trail to the vehicle ECU via the network, and wherein the method further comprises transmitting the images received by the vehicle ECU to either the remote operating device or GUI for viewing on the screen.
